
ULTRA-MASTER SYSTEM BIBLE

Version 8 — Complete Reference

Smarter Poker — Club Arena
Server-Authoritative Poker Engine Specification

Chapters 1-11 + Appendices A-F
March 2026

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CHAPTER 1: MASTER LAWS

Law 1.1 — SINGLE PENDING ACTION

At any moment during a hand, exactly ONE player has the right to act. No other player may submit actions until the current player's action is processed and the turn advances.

Requirements:

- **1.1.1:** Server maintains `currentPlayerSeat` — only this seat's actions are accepted
- **1.1.2:** All other action submissions are rejected with clear error
- **1.1.3:** Pre-actions are queued but NOT executed until it becomes that player's turn
- **1.1.4:** No parallel action processing — actions are serialized

Law 1.2 — HARD BLOCK LAW

The game MUST NOT advance to the next step until the current step is fully complete. No action, no deal, no broadcast may occur out of order.

- **1.2.1:** Action must be validated before execution
- **1.2.2:** Execution must complete before state broadcast
- **1.2.3:** Broadcast must confirm before next turn begins
- **1.2.4:** Timer starts only AFTER broadcast confirms turn change
- **1.2.5:** No fire-and-forget — every step awaits completion

Law 1.3 — ORDER OF OPERATIONS

Every player action follows this EXACT sequence (20 steps):

1. Player clicks UI button
2. Client sends action to server
3. Server validates identity (auth check)
4. Server validates turn (is it this player's turn?)
5. Server validates action legality (can they check/bet/raise?)
6. Server validates amount (min raise, max raise, stack limits)
7. Server validates timing (within timer + grace period)
8. Server checks for duplicate submission
9. Server executes action on `HandController`
10. Server updates game state (pot, bets, stacks, player status)
11. Server calculates new available actions for next player
12. Server determines next player to act
13. Server starts next player's timer
14. Server broadcasts authoritative state to all clients
15. Client receives broadcast
16. Client updates UI state from server data

17. Client shows popup/overlay for the action
18. Client plays animation (chips moving, cards dealing)
19. Client plays sound effect
20. Client triggers haptic feedback

Law 1.4 — TRUTH LAW

There is ONE source of truth: the server. Client state is a REFLECTION of server state, never authoritative.

- **1.4.1:** Server state is the canonical game state
- **1.4.2:** Client derives ALL display state from server broadcasts
- **1.4.3:** If client and server disagree, server wins — always
- **1.4.4:** StateVerifier runs between hands to ensure chip conservation
- **1.4.5:** No client-side game logic that modifies canonical state

Law 1.5 — FAIRNESS LAW

Every player receives equal treatment: correct turn order, legal action sets, timer rights, side-pot eligibility, and card security.

- **1.5.1:** Turn order follows standard poker position rules
- **1.5.2:** Available actions are calculated identically for all players
- **1.5.3:** Timer duration is the same for all players (unless time bank differs)
- **1.5.4:** Side pot eligibility is calculated correctly for all-in players
- **1.5.5:** No player can see another player's hole cards (anti-god-mode)
- **1.5.6:** Validation errors return error messages, NEVER auto-fold

Law 1.6 — NO-AMBIGUITY LAW

Every state in the system has explicit entry conditions, exit conditions, and failure conditions. No implicit state transitions.

Law 1.7 — DISCONNECT LAW

When a player disconnects, their action timer continues running. The game does NOT pause for disconnected players.

- **1.7.1:** Disconnect detected via heartbeat (server-side)
- **1.7.2:** Action timer keeps counting during disconnect
- **1.7.3:** If timer expires during disconnect: auto-fold (or auto-check if possible)
- **1.7.4:** preferCheckOverFold: if player can check, auto-check instead of auto-fold
- **1.7.5:** Reconnect grace period: 5 seconds after reconnect before next auto-action
- **1.7.6:** maxConsecutiveTimeouts: after N consecutive timeouts (default 3), auto-sit-out

Law 1.8 — FOLD FINALITY LAW

Once a player folds, their `is_folded` flag is TRUE for the remainder of the hand. No undo. No take-backs.

Law 1.9 — SETTLEMENT LAW

Hand settlement follows this EXACT sequence:

1. Lock table (no new actions accepted)
2. Calculate side pots from contributions
3. Evaluate all active players' hands (variant-aware)
4. Determine winners per pot (including hi-lo split)
5. Calculate rake (percentage with cap, no-flop-no-drop)
6. Distribute winnings (integer-cents arithmetic)
7. Update player stacks
8. Persist results to database (atomic transaction)
9. Update leaderboards
10. Trigger achievements/daily challenges
11. Calculate VIP points earned
12. Calculate rakeback
13. Log complete hand history
14. Broadcast final state (with showdown cards)
15. Unlock table

Law 1.10 — VISUAL TRUTH LAW

Every visual element (popup, overlay, badge, animation) must correspond to an actual game event. No visual without an event, no event without a visual.

Law 1.11 — AUDIO TRUTH LAW

Every sound effect corresponds to exactly one game event. No sound for non-events, no missing sound for events.

Law 1.12 — HAPTIC TRUTH LAW

Haptic feedback intensity matches event significance: light for check/fold, medium for bet/raise, heavy for all-in/win.

Law 1.13 — PRIORITY STACK

Server > Database > Client. In any conflict, higher priority wins.

Law 1.14 — SCOPE

This Bible covers: Cash games, Sit-and-Go, Multi-Table Tournaments, Spin & Go, all Hold'em variants, all Omaha variants, Short Deck, and OFC.

Law 1.15 — CASH/TOURNAMENT/VARIANT BRANCHING

Rules that differ between cash and tournament (rebuys, blind increases, elimination) are explicitly branched. Rules that differ between variants (cards dealt, hand evaluation, hi-lo) are explicitly branched.

CHAPTER 2: OBJECT SCHEMAS

2.1 — Table Object

id, club_id, tournament_id, name, game_type ('cash'|'tournament'), game_variant, stakes, small_blind, big_blind, ante, big_blind_ante, max_players, min_players, status, settings, created_at, updated_at

2.2 — Table Settings Object

straddle_enabled, straddle_type ('utg'|'mississippi'), max_straddles, run_it_twice_enabled, bomb_pot_enabled, bomb_pot_frequency, bomb_pot_ante_multiplier, time_bank_enabled, time_bank_seconds, time_bank_max_uses, action_time_seconds, insurance_enabled, ante_enabled, ante_amount, big_blind_ante_enabled, disconnect_timeout_seconds, max_consecutive_timeouts, prefer_check_over_fold, auto_muck_enabled, show_hand_enabled

2.3 — Seat/Player Object

seat, user_id, username, stack, bet, totalInvested, cards, is_folded, is_all_in, is_sitting_out, is_disconnected, time_bank_remaining, time_bank_uses_remaining, position, avatar_url, is_horse

2.4 — Hand State Object

table_id, hand_number, pot, community_cards, current_bet, current_player, dealer_seat, stage, min_raise, last_raise, turn_start_time_ms, turn_duration_ms, players[], pots[], action_history[]

2.5 — Action Record

seat, userId, action, amount, timestamp, stage

2.6 — Pot Object

amount, eligible (user_id[])

2.7 — Winner Object

userId, amount, potIndex, hand (evaluated hand description)

2.8 — Hand Config

tableId, handNumber, gameVariant, smallBlind, bigBlind, ante, bigBlindAnte, rakeConfig, bombPot, straddles, ritEnabled, insuranceEnabled

2.9 — Rake Config

percent, cap, noFlopNoDrop, playerCountCaps[], timedRake (alternative)

2.10-2.18

Additional schemas for tournaments, achievements, hand history layers (4-tier: raw, structured, display, export)

CHAPTER 3: STATE MACHINES

3.1 — Table State Machine

EMPTY	-> WAITING: first player sits
WAITING	-> SEATING: second player sits (buy-in phase)
SEATING	-> RUNNING: both bought in, hand starts
RUNNING	-> RUNNING: hand completes, next hand starts
RUNNING	-> PAUSED: admin pause, break, or hand-for-hand
PAUSED	-> RUNNING: resume
RUNNING	-> WAITING: player count drops below 2
RUNNING	-> CLOSING: admin close or all leave
CLOSING	-> CLOSED: cleanup complete

3.2 — Hand State Machine

IDLE -> POSTING_BLINDS -> DEALING_HOLE_CARDS -> PREFLOP_BETTING -> DEALING_FLOP -> FLOP_BETTING -> DEALING_TURN -> TURN_BETTING -> DEALING_RIVER -> RIVER_BETTING -> SHOWDOWN -> SETTLEMENT -> CLEANUP -> IDLE

Each state has: entry conditions, actions, exit conditions, failure conditions.

3.3 — Turn State Machine

WAITING	-> TIMER_RUNNING: broadcast received, this player's turn
TIMER_RUNNING	-> TIME_BANK_ACTIVE: primary timer expires, time bank available
TIMER_RUNNING	-> EXPIRED: primary timer expires, no time bank
TIME_BANK_ACTIVE	-> EXPIRED: time bank expires
TIMER_RUNNING	-> ACTION_RECEIVED: player submits action within time
ACTION_RECEIVED	-> PROCESSING: server validates + executes
PROCESSING	-> COMPLETE: action applied, next turn begins
EXPIRED	-> COMPLETE: auto-fold/check applied

3.4 — Disconnect State Machine

CONNECTED	-> HEARTBEAT_MISSED: no heartbeat for N seconds
HEARTBEAT_MISSED	-> DISCONNECTED: no heartbeat for disconnect_timeout_seconds
DISCONNECTED	-> RECONNECTING: heartbeat received again
RECONNECTING	-> RECONNECTED: grace period complete
RECONNECTED	-> CONNECTED: full state sync complete

CHAPTER 4: OPERATIONAL PROCEDURES

4.1 — Hand Start Procedure

1. Select dealer (rotate from previous hand)
2. Determine positions (BTN, SB, BB, UTG, etc.)
3. Post blinds (handle heads-up: dealer=SB)
4. Post antes (if enabled — traditional or BBA)
5. Post straddles (if enabled — UTG or Mississippi)
6. Shuffle deck (crypto-random)
7. Deal hole cards (2 for Hold'em, 4-6 for Omaha variants)
8. Deliver hole cards securely (per-player, not broadcast)
9. Set first player to act (UTG preflop, or SB postflop in heads-up)
10. Start action timer
11. Broadcast hand state

4.2 — Blind Posting

- **Heads-up:** Dealer posts SB, other player posts BB
- **3+ players:** Player left of dealer posts SB, next left posts BB
- **Short blind:** Post what they can, mark all-in
- **Dead blind:** Player returning from sit-out posts both SB+BB, SB is dead

4.3 — Ante Handling

- **Traditional ante:** Every player posts ante before cards
- **Big Blind Ante (BBA):** BB posts ante for entire table
- Ante goes into pot before blinds
- If player can't cover ante: post what they can

4.4 — Straddle Handling

- **UTG straddle:** UTG posts 2x BB before cards, action starts left of straddler
- **Mississippi straddle:** Any position can straddle, last straddler acts last preflop
- **Re-straddle:** Next player can straddle 2x previous straddle (up to max_straddles)
- Straddle is live — straddler can raise when action comes back

4.5 — Card Dealing

- **Shuffle:** crypto-random (crypto.getRandomValues or equivalent)
- **Hold'em:** 2 cards per player
- **PLO/PLO4:** 4 cards per player | **PLO5:** 5 cards | **PLO6:** 6 cards
- **Short Deck:** Remove 2s through 5s before shuffle

- Deal one card at a time, clockwise from SB

4.6 — Hole Card Security (Anti-God-Mode)

- Server deals cards and stores per-player
- Each player receives ONLY their own cards via secure per-player channel (RPC, not broadcast)
- Broadcast state NEVER includes other players' hole cards
- At showdown: reveal only non-folded players' cards (or winners only if auto-muck)

4.7-4.8 — Betting Round Flow

1. Determine first player to act (preflop: left of BB; postflop: first active left of dealer)
2. Player acts: fold, check, call, bet, raise, or all-in
3. Advance to next active non-all-in player
4. Repeat until betting round complete

Round complete when: All non-all-in players have acted since last aggressive action AND all bets are equal, OR only one player remains.

4.9-4.14 — Action Validation Rules

Action	Legal When	Amount
Fold	Always legal	N/A (auto-check if toCall=0)
Check	toCall = 0	0
Call	toCall > 0	min(toCall, playerStack)
Bet	currentBet = 0	>= minBet (usually BB)
Raise	currentBet > 0	>= currentBet + lastRaise
All-in	Player has chips	Remaining stack

- **Min raise:** Raise must be at least the size of the last raise (or BB if first raise)
- **Max raise:** No-limit = entire stack; Pot-limit = current pot + call amount
- **Short all-in:** If player goes all-in for less than a full raise, it does NOT reopen betting

4.15 — Pre-Action System

Pre-Action	Behavior
auto_fold	Fold when my turn comes
auto_check_fold	Check if possible, otherwise fold
auto_check	Check if possible (invalid if there's a bet)
auto_call	Call current bet when my turn comes
auto_call_any	Call any bet (including raises after setting)

- Pre-actions are evaluated when the turn reaches the player
- If game state changed (someone raised), pre-action may be invalidated
- Pre-action is ALWAYS cleared after evaluation (valid or not)

- Pre-action execution follows same validation as manual actions

4.16-4.18 — Stage Progression

- **Preflop -> Flop:** Deal 3 community cards, reset bets, first player left of dealer
- **Flop -> Turn:** Deal 1 community card, reset bets
- **Turn -> River:** Deal 1 community card, reset bets
- **River -> Showdown:** If 2+ players remain after betting
- **All-in runout:** Deal remaining community cards automatically (with pause for RIT/insurance)

4.19 — Insurance

1. Calculate equity for each player
2. Offer insurance to the player with the best hand
3. Player can buy insurance on specific outs
4. If bought: insurance payout calculated regardless of board runout
5. Time limit to accept insurance offer

4.20 — Run It Twice (RIT)

1. Offer RIT to both players (exactly 2 all-in)
2. Both must accept (either can decline)
3. If accepted: deal two separate boards from the remaining deck
4. Each board determines half the pot
5. Rake applies once (not per board); 3x boards also supported (1/3 each)
6. Time limit to accept RIT offer

4.21 — Showdown Rules

- Last aggressor shows first
- If no aggressor (everyone checked river): first player left of dealer shows first
- Players can muck (hide cards) — but winner must show
- **Auto-muck:** If enabled, losing hands are automatically mucked
- **Show hand:** Player can choose to show even when not required
- Showdown order proceeds clockwise from the first shower

4.22 — Bomb Pot

- All players post ante (multiplier x BB)
- Skip preflop betting — deal directly to flop
- Normal betting from flop onward
- Frequency: every N hands (configurable)
- PLO bomb pots: deal 4-6 cards, bet from flop

CHAPTER 5: UI / POPUP / ANIMATION / SOUND / HAPTIC DOCTRINE

5.1 — Popup Doctrine

Every significant game event triggers a popup/overlay:

- Hand start: position badges appear
- Blind posting: chip animation from player to pot
- Card dealing: card flip animation
- Player action: action label popup (FOLD, CHECK, CALL \$X, RAISE \$X, ALL-IN)
- Community cards: card slide animation with stage label
- Showdown: card reveal animation with hand rank label
- Winner: chip slide animation from pot to winner with amount label
- Insurance/RIT: modal overlay with accept/decline buttons

5.2 — Animation Sequence

Animations are SEQUENTIAL, not simultaneous:

1. Action label appears (200ms)
2. Chip animation plays (300ms)
3. Pot updates (100ms)
4. Turn indicator moves (200ms)

5.3 — Sound Doctrine

Event	Sound
fold	Soft card toss
check	Tap/knock
call	Chip clink
bet/raise	Chip stack (louder for larger amounts)
all_in	Dramatic chip slide + bell
deal	Card slide
community cards	Card flip
showdown	Dramatic reveal
winner	Celebration (confetti for large pots)
timer warning	Tick in last 5 seconds
time bank	Bank activation
disconnect	Subtle offline indicator

5.4 — Haptic Doctrine

Event	Intensity
fold/check	Light haptic
call	Light haptic
bet/raise	Medium haptic
all_in	Heavy haptic
your turn	Medium haptic
you win	Heavy celebration haptic
timer warning (3s)	Quick pulse haptic

CHAPTER 6: TIMER SYSTEM

6.1 — Action Timer

- Server-authoritative, **deadline-based** (not setTimeout)
- Configurable per table: action_time_seconds (default 15)
- Timer starts when server broadcasts TURN_CHANGE
- Timer displayed on client as countdown
- Grace period: 2 seconds after timer reaches 0 (for network latency)
- On expiry: auto-fold (or auto-check if toCall=0 and preferCheckOverFold=true)

6.2 — Time Bank

- Per-hand limit: 2 activations max (1 auto + 1 manual)
- **Auto-activate:** When primary timer expires and time bank available
- **Manual activate:** Player clicks 'Time Bank' button during their turn
- Each activation adds configurable seconds (default 15s per use)
- **Pool model:** Total seconds available per session, depletes per use
- **Refill:** Per orbit or per session (configurable)

6.3 — Disconnect Timer

- Heartbeat interval: server pings every 3-5 seconds
- Disconnect detected: no heartbeat for disconnect_timeout_seconds (default 30s)
- During disconnect: action timer keeps running
- On disconnect + timer expiry: auto-fold/check
- maxConsecutiveTimeouts (default 3): after N consecutive timeouts, auto-sit-out
- Reconnect grace: 5 seconds after heartbeat resumes before next auto-action

CHAPTER 7: EDGE CASES (200+ Test Scenarios)

ID	Scenario
7.1	Heads-up blind posting (dealer=SB)
7.2	Short blind (can't cover, all-in immediately)
7.3	All-in for less than minimum raise (doesn't reopen betting)
7.4	Side pot calculation with 3+ all-in players at different amounts
7.5	Split pot (identical hands)
7.6	Hi-Lo split with no qualifying low
7.7	Hi-Lo split with odd chip (goes to high winner)
7.8	Run-it-twice with different winners on each board
7.9	Disconnect during all-in runout
7.10	Bomb pot with player who can't cover ante
7.11	Straddle when next player can't cover
7.12	Player sits out during a hand (wait until next hand)
7.13	Player leaves table during a hand (process at hand end)
7.14	Tournament elimination (last chip to pot, player exits)
7.15	Final table bubble (hand-for-hand play)
7.16	Simultaneous disconnects at same table
7.17	Server crash recovery (reload state from DB, resume)
7.18	Rake calculation with no flop (no-flop-no-drop)
7.19	Rake cap per player count
7.20	Mixed game rotation (change variant every orbit)

CHAPTER 8: EXTENSIBILITY

8.1 — New Game Variants

Adding a new variant requires:

- New evaluator function
- Cards-per-player configuration
- Blind/ante/straddle rules if different
- UI card layout if different

8.2 — New Tournament Types

Adding a new tournament type requires:

- Blind structure definition
- Payout structure
- Elimination/rebuy/addon rules
- Timer/break configuration

CHAPTER 9: WORLD-CLASS EXCELLENCE

9.1 — Performance

Metric	Target
Action processing	< 50ms server-side
Broadcast latency	< 100ms to all clients
UI update	< 16ms (60fps target)
Hand throughput	30+ hands/hour at full pace

9.2 — Reliability

- Zero tolerance for chip leaks (StateVerifier between every hand)
- Zero tolerance for card exposure
- Graceful degradation on network issues
- Auto-recovery from crashes

9.3 — Security

- All game logic server-side
- Per-player card provisioning
- Rate limiting on action submissions

- Auth validation on every request

CHAPTER 10: ANIMATION STANDARDS

10.1 — Card Animations

- **Deal:** Cards slide from deck to player positions
- **Community:** Cards slide to center, flip face-up
- **Showdown:** Cards flip to reveal hand

10.2 — Chip Animations

- **Bet/raise:** Chips slide from player to pot
- **Win:** Chips slide from pot to winner

10.3 — Timing

- All animations are sequential, not overlapping
- Total animation budget per action: < 800ms
- **Skip animations option** for speed players

CHAPTER 11: TABLE SETTINGS & THEME CUSTOMIZATION

11.1 — Table Settings (User Preferences)

Every table setting toggle MUST exist in TWO locations: (1) Table Settings panel — accessible from the table view gear icon, and (2) Club Arena hamburger menu — accessible from the existing side navigation. Both locations read/write the SAME `user_table_settings` row in Supabase. Changes persist across sessions.

11.1.1 — Required Toggle Settings

Setting	DB Column	Default	Description
Highlight Active Players	<code>highlight_active_players</code>	true	Highlight the currently-acting player's seat
Show Avatars	<code>show_avatars</code>	true	Display player avatar images at seats
Show Badges	<code>show_badges</code>	false	Display VIP/achievement badges at seats
Cards Pre-Sort	<code>cards_pre_sort</code>	true	Auto-sort hole cards by rank (high to low)
Gestures	<code>gestures_enabled</code>	false	Enable swipe/drag gesture controls
Card Slide	<code>card_slide</code>	false	Enable card peek/slide reveal animation
Show Stack in BBs	<code>show_stack_in_bb</code>	false	Display stacks as BB count
Auto Time Bank	<code>auto_time_bank</code>	false	Auto-activate time bank when timer expires
Enhanced View	<code>enhanced_view</code>	false	Enable enhanced visual effects
Voice Message	<code>voice_message</code>	true	Enable voice chat at table
Text Message	<code>text_message</code>	true	Enable text chat at table
Emoji	<code>emoji_enabled</code>	true	Enable emoji reactions/throwables

11.1.2 — Persistence

- All settings stored in `user_table_settings` table (keyed by `user_id`)
- Client reads on `TablePage` mount via `supabase.from('user_table_settings').select()`
- Changes saved immediately on toggle via `supabase.from('user_table_settings').upsert()`
- Settings apply to ALL tables the user joins (not per-table)

11.2 — Theme Settings (Per-Game-Type Customization)

Theme customization is accessed via the 'Theme Setting' row in Table Settings. Themes are customizable PER GAME TYPE — each game type can have independent theme selections.

11.2.1 — Game Types for Theme Customization

ALL | NLH | FLH | 6+ | PLO | FLO | OFC | MIXED | MTT | SNG

11.2.2 — Theme Categories (5 Tabs)

Tab	Category	Free (2)	VIP (3)
1	Themes (full presets)	Default dark, Classic brown	Neon blue hexagonal, Rustic wood/leather, Casino green felt
2	Table (felt/surface)	Dark felt, Brown felt	Neon blue, Red leather, Green casino felt
3	Button (dealer)	Red 'D' gear, Gray 'D' gear	Blue crystal 'D', Gold star badge, Sports themed
4	Background (room)	Diamond pattern, Stone/concrete	Galaxy/space nebula, Hardwood floor, Teal tile pattern
5	Cards (face design)	Standard red, Standard blue	Premium gold, Premium black, Platinum

11.2.3 — VIP Gating (Single Tier)

- **Free items:** Available to all users (2 per category)
- **VIP items:** Locked with 'VIP' badge overlay — requires active VIP subscription
- Client checks user's `is_vip` from profiles table
- Locked items show padlock overlay; tapping shows VIP upgrade prompt

11.2.4 — Theme Persistence

- Stored in `user_theme_settings` table
- Schema: `user_id, game_type, theme_id, table_id, button_id, background_id, cards_id`
- Each row = one game type's theme selections for one user
- 'ALL' game type = default fallback if no per-game-type override exists
- Reset button: reverts all selections to Free defaults for current game type
- Confirm button: saves selections and closes Theme Settings modal

11.2.5 — Asset Requirements

- Each theme asset needs: thumbnail image (for selector grid), full asset (for rendering)
- Thumbnails: ~100x80px for grid display
- Full assets: table felt texture, button sprite, background image, card face sprites
- All assets served from `smarter.poker/hub/club-arena/assets/themes/`

11.3 — Implementation Notes

- Table Settings component should be a reusable `<TableSettingsPanel />` that renders in both the table gear menu and the hamburger nav
- Theme Settings is a modal/page with 5-tab layout, game type dropdown at top
- Both require Supabase RPC or direct table access for persistence
- VIP gating requires reading the user's current VIP level and comparing against asset tier requirements

APPENDICES

Appendix A: Rake Chart

- Standard rake: 10% of pot with tier-based cap
- No-flop-no-drop: if hand doesn't reach flop, no rake
- Caps by blind level (see code for exact chart)

Appendix B: Position Names

Players	Positions
2	BTN/SB, BB
3	BTN/SB, BB, UTG
4	BTN, SB, BB, UTG
5	BTN, SB, BB, UTG, CO
6	BTN, SB, BB, UTG, MP, CO
7	BTN, SB, BB, UTG, UTG+1, MP, CO
8	BTN, SB, BB, UTG, UTG+1, MP, MP+1, CO
9	BTN, SB, BB, UTG, UTG+1, UTG+2, MP, HJ, CO

Appendix C: Hand Rankings (Hold'em)

Royal Flush > Straight Flush > Four of a Kind > Full House > Flush > Straight > Three of a Kind > Two Pair > One Pair > High Card

Appendix D: Hand Rankings (Short Deck)

- Flush beats Full House (harder to make with fewer cards)
- A-6-7-8-9 is the lowest straight (ace plays low)

Appendix E: Animation Physics

- Chip trajectories follow parabolic arcs
- Card deals follow bezier curves
- Winner celebration uses particle system

Appendix F: Sound Library Requirements

Each sound has: file reference, duration, volume level, priority (can interrupt lower priority).

END OF DOCUMENT
ULTRA-MASTER SYSTEM BIBLE v8 — Smarter Poker Club Arena